

TrES-3b

R-2826

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140803 · created 2026-09-05T23:07:02+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456872.735 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.057
Transit depth (Rp/Rs)^2	3.77 +/- 2.2 [%]
Orbital Inclination (inc)	77.44 +/- 2.81
Airmass coefficient 1 (a1)	11.18 +/- 7.13
Airmass coefficient 2 (a2)	-1.88 +/- 0.9
Scatter in the residuals of the lightcurve fit is	57.4 %
Best Comparison Star	#2 - [174, 36]
Optimal Aperture	2.29
Optimal Annulus	4.25
Transit Duration (day)	0.011 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.057 vs 0.16309 (deviation 0.54σ)
Duration fit vs published	0.011 d vs 0.0567 d (deviation 2.17σ)
Mid-time O–C	-2.6 min
Depth SNR	3.4
Residual scatter	57.4 %

Light curve

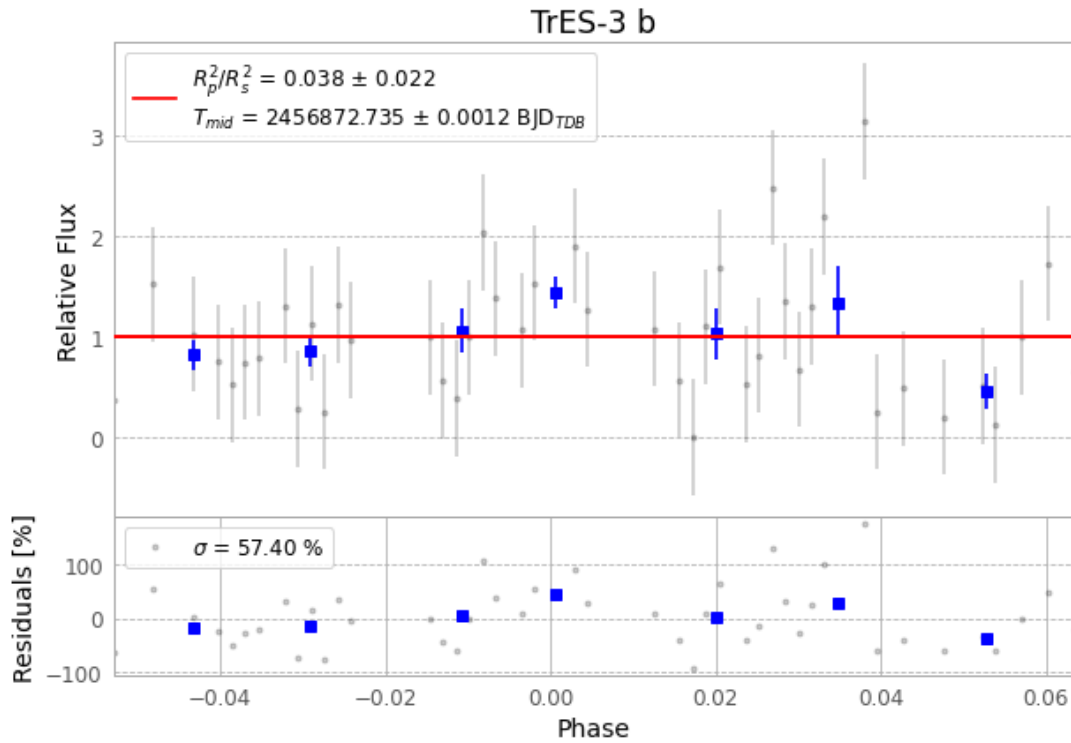


Figure 1 — FinalLightCurve_TrES-3 b_2014-08-02.png (SHA-256 a9d39d2403048646...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3af3105b733878d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

74 FITS frames (49 MB), TRES-3140803035413.FITS → TRES-3140803073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140803.log (84623410b8e3...), FinalLightCurve_TrES-3 b_2014-08-02.png (a9d39d240304...), FinalLightCurve_TrES-3 b_2014-08-02.csv (bff2a6c60508...)

Compute resources

Wall time 293.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:07Z.